

SOFTWARE ENGINEERING Common to CSE&IT

III Semester

Course Code:201CS3T04

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Course Outcomes:**At the end of the Course, Student will be able to:**

- CO1:** Demonstrate the fundamental concepts of software engineering
CO2: Compare Process development models for efficient software development.
CO3: Describe the activities of Software Project Management for any application.
CO4: Create the software process artifacts (SRS document, Design diagrams, test cases) for any application.
CO5: Analyze quality and reliability of software.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	1	-	-	-	-	2	2	2	1
CO3	2	1	1	-	-	-	-	-	-	-	2	1
CO4	3	2	2	1	-	-	-	-	2	2	2	1
CO5	1	2	-	-	-	-	-	-	-	-	-	-

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	1	-
CO2	2	-
CO3	1	1
CO4	2	2
CO5	1	1

Unit - I

Introduction to Software Engineering: The Nature of Software, The Unique Nature of WebApps, Software Engineering, The Software Process, Software Engineering Practice, Software Myths.

Software Process: Software Process, Process Classification, Phased Development Life Cycle, Software Development Process Models – Waterfall Model, Iterative Waterfall Model, Prototype Model, Incremental Model, Spiral Model, Agile Process Model and RUP process Model.

Case Study: Survey on different process models including.

- i. Advantages and Disadvantages of the models.
- ii. Applicability of the model.
- iii. Projects developed using various models.

Unit – II

Software Project Management: Project Management Essentials, What is Project Management, Software Configuration Management, Risk management.

Project Planning and Estimation: Project Planning Activities, Software Metrics and Measurements, Project Size Estimation, Effort Estimation Techniques.

Case Study: Estimate the effort of the software development using Functional Points and COCOMO Model for any real time problem.

Unit – III

Requirements Engineering: Software Requirements, Requirements Engineering Process, Requirements Elicitation and Analysis, Requirements Specification, Requirements Validation, Requirements Management.

Case Study: Create a SRS document for any one of the following Software Projects.

1. Course Registration System
2. Students Marks Analyzing System
3. Online Ticket Reservation System
4. Stock Maintenance

Unit – IV

Software Design: Software Design Process, Characteristics of Good Software Design, Design Principles, Modular Design, Software Architecture, Design Methodologies,

Implementation: Coding Principles, Coding Process, Code Verification, and Code Documentation.

Case Study: Construct the DFD and CFD for any one of the following Software Projects.

1. Airline Reservation System
2. Students Marks Analyzing System
3. ATM System
4. Library Management System

Unit – V

Software Testing: Testing Fundamentals, Test Planning, Black-Box Testing, WhiteBox Testing, Levels of Testing, Usability Testing, Regression Testing, Debugging Approaches.

Software Quality and Reliability: Software Quality factors, Verification & Validation, Software Quality Assurance, The Capability Maturity Model, and Software Reliability.

Case Study: Design the test cases for any one of the following real time scenarios using White Box & Black Box Testing Techniques.

1. E-Commerce application (Flipkart, Amazon)
2. Mobile Application

Text Books:

1. Software Engineering – Concepts and Practices: Ugrasen Suman, Cengage Learning.
2. Software Engineering: A practitioner's approach, Roger S. Pressman, McGrawHill.

Reference Books:

1. Software Engineering, Ian Sommerville, Ninth Edition, Pearson
2. Software Engineering, A Precise Approach, Pankaj Jalote, Wiley India
3. Fundamentals of Software Engineering, Rajib Mall, Prentice Hall India.
4. Software Engineering: A Primer, Waman S Jawadekar, Tata McGraw-Hill.
5. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

Web Links:

1. https://www.tutorialspoint.com/software_engineering/
2. <http://nptel.ac.in/courses/106/105/106105182/>
3. <https://www.coursera.org/learn/software-processes-and-agile-practices>
4. <http://www.geeksforgeeks.org/software-engineering-gq/>
5. <https://www.coursera.org/browse/computer-science/software-development>